

Mingfeng Zhong

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Education

Rochester Institute of Technology (RIT)

Expected May 2028

- **Major:** Software Engineering B.S | Game Design & Development Minor | GPA 3.25
- **Achievements:** Certified W3Schools JavaScript Developer, Dean's List Fall & Spring 2023-2024, RIT Presidential Scholarship

Experience

Rochester Institute of Technology

May, 2026 – Current

Flight Software Researcher

- Communicated with **the Federal Aviation Administration** and project stakeholders to define requirements for an open-source **Java Dockerized** flight-data management service analyzing flight and aircraft risks
- Implemented dynamic obstacle loading and aircraft-specific collision-risk estimation for **600,000+** documented obstacles, expanding JUnit test coverage across diverse flight-collision scenarios to validate edge cases and system reliability
- Flagged and cleaned **4,000+** lines of irregular flight-maintenance logs using Python, improving the quality of data used by the flight-analysis service

Midwood Science at Midwood High School

Apr. 2022 – Apr. 2023

JATOS Advisor (Web Developer)

- Developed interactive web applications and scientific experiments in **Agile** sprints, utilizing JATOS to provide non-technical administrators with accessible interfaces for managing project data and configurations
- Deployed **AWS** EC2 proxy infrastructure to secure and encrypt user connections, enabling remote access to web-based projects
- Transitioned multiple in-person scientific experiments to remote web applications, enabling participants to conduct experiments remotely and increasing overall productivity by **20%**

Projects

Embedded MIDI Music Player

Sep. 2025 – Dec. 2025

- Engineered a MIDI music player on an **ARM Cortex** microcontroller using **C**, designing and optimizing a MIDI parsing algorithm by analyzing file structures and technical documentation
- Streamlined code execution through non-blocking polling, increasing software speed and responsiveness by **70%**

Pharmaceutical Employee Portal

Aug. 2025 – Dec. 2025

- Designed and implemented a full stack React application simulating internal employee operations for a medical company, structuring employee records in **MongoDB** to support rapid iteration and adaptable feature development
- Co-led a team of 5 through code integration and deployment, deploying the application to an **Ubuntu** server using **Apache2**

Leadership

Chinese Conversation Table Club at RIT

Jan. 2026 – Current

Secretary & Web Developer

- Designed and facilitated 5+ cultural immersion events featuring traditional Chinese folk art and cultural practices
- Redesigned the club website and expanded social media channels in collaboration with other Asian clubs

Society of Software Engineers at RIT

Aug. 2025 – Dec. 2025

Student Tutor

- Performed in-depth code analysis and debugging across 3 student-developed software projects, guiding students through identifying and resolving issues without directly providing solutions
- Curated study guides and delivered customized assessments to 10+ students on core software engineering concepts, emphasizing conceptual understanding and practical application

Skills

Languages: Java, Python, C/C++, JavaScript/TypeScript, SQL

Frameworks: React, Node.js, Spring Boot, Flask, JUnit

Tools & Databases: Git, Docker, AWS, MySQL, PostgreSQL, MongoDB